

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool : Analytical Problem Solving (High)
Assessment Tool Weightage: 35%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Apply relational algebra operations (σ , π , $\cup$, $-$, $\times$, $\bowtie$) on the given Employee and Department relations to retrieve required records.	BL3 – Apply	5
2	Write SQL DDL statements to create STUDENT(SID, Name, DOB, DeptID) and DEPARTMENT(DeptID, DeptName) tables with all appropriate constraints.	BL3 – Apply	5
3	Formulate SQL queries using GROUP BY, HAVING, and aggregate functions (COUNT, SUM, AVG, MAX, MIN) on a given Sales dataset.	BL3 – Apply	5
4	Write a correlated sub-query to find employees earning more than the average salary of their own department.	BL3 – Apply	5
5	Apply all types of JOIN operations (INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, CROSS) on Employee and Project tables.	BL3 – Apply	5

6	Create a view 'DeptSalaryView' that displays department name, total employees, and average salary. Write a query on this view.	BL6 – Create	5
7	Construct a relational algebra expression for the following: Find names of students who are enrolled in all courses offered.	BL6 – Create	5
8	Write SQL DML statements (INSERT, UPDATE, DELETE) with integrity constraint enforcement for a given scenario.	BL3 – Apply	5
9	Apply tuple relational calculus to express a query: Find all employees who work in the 'IT' department and earn more than 50000.	BL3 – Apply	5
10	Design a relational schema for an Airline Reservation System and write 5 SQL queries demonstrating joins, sub-queries, and views.	BL6 – Create	5

Code of Conduct:

I Ns avesh huusain (192525241) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Ns avesh huusain

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Criteria	Weightage (%)	Level 4 (Exemplary) (85-100%)	Level 3 (Proficient) (70-84%)	Level 2 (Developing) (50-69%)	Level 1 (Beginning) (0-49%)	Marks Awarded (Out of)
1. Problem Understanding	20%	☐	☐	☐	☐	_____
2. Method Selection	20%	☐	☐	☐	☐	_____
3. Solution Process	25%	☐	☐	☐	☐	_____
4. Accuracy of Calculation	20%	☐	☐	☐	☐	_____
5. Interpretation of Results	15%	☐	☐	☐	☐	_____
Total Marks	100%					_____ / 50

Q1) Relational Algebra is a **procedural query language** used in Relational Database Management Systems (RDBMS). It provides a collection of operations to retrieve and manipulate data stored in relational tables. It forms the theoretical foundation of SQL and helps in processing database queries efficiently. Each operation performs a specific task and produces another

relation as the output.

Key Points

1. Relational Algebra operates on one or more relations (tables).
2. It is the foundation of SQL query processing.
3. **Selection (σ)** retrieves rows that satisfy a specified condition.
4. **Projection (π)** retrieves only the required columns from a table.
5. **Union ($\cup$)** combines tuples from two compatible relations without duplicates.
6. **Set Difference ($-$)** returns tuples that are present in one relation but not in another.
7. **Cartesian Product ($\times$)** combines every row of one relation with every row of another relation.
8. **Join ($\bowtie$)** combines records from two tables based on a common attribute.
9. These operations help retrieve meaningful information from databases.
10. Relational Algebra improves query optimization and database performance.

Q2) Introduction

Data Definition Language (DDL) is a part of SQL used to define and manage the structure of database objects such as tables. In a student database, the STUDENT and DEPARTMENT tables are related through the Department ID. Appropriate constraints are applied to maintain data integrity, accuracy, and consistency within the database.

Key Points

1. DDL is used to create and define database tables.
2. The STUDENT table stores Student ID, Name, Date of Birth, and Department ID.
3. The DEPARTMENT table stores Department ID and Department Name.
4. Primary Key uniquely identifies each record in a table.
5. Foreign Key creates a relationship between the STUDENT and DEPARTMENT tables.
6. NOT NULL ensures mandatory fields are not left empty.
7. UNIQUE prevents duplicate values in important columns.
8. Constraints improve data integrity and consistency.
9. Proper table design reduces redundancy and maintains relationships.
10. Well-designed tables improve database performance and reliability.

Q3) Introduction

GROUP BY, HAVING, and Aggregate Functions are important SQL features used to organize and analyze data in a database. Aggregate functions perform calculations on a group of records and return a single result. The GROUP BY clause groups rows with the same values, while the HAVING clause filters grouped data based on a specified condition. These features are widely used in sales analysis, employee management, and report generation.

Key Points

1. Aggregate functions are used to perform calculations on multiple records.
2. COUNT() returns the total number of records in a table or group.
3. SUM() calculates the total value of a numeric column, such as total sales.
4. AVG() calculates the average value of a numeric column.
5. MAX() returns the highest value from a column.
6. MIN() returns the lowest value from a column.
7. The GROUP BY clause groups records that have the same values into one result.
8. The HAVING clause filters grouped records after the GROUP BY operation.
9. These SQL features are commonly used to generate department-wise reports, monthly sales reports, and employee salary summaries.
10. They improve data analysis, simplify report generation, and support better business decision-making.

Applications

- Sales analysis and reporting.
- Employee salary analysis.
- Student performance reports.
- Inventory management.
- Business decision-making.

Advantages

- Summarizes large amounts of data.
- Makes reports easy to understand.

- Improves query efficiency.
- Supports accurate data analysis.
- Helps organizations make informed decisions.

Q4)

Introduction

A correlated subquery is a type of SQL subquery that depends on the outer query for its execution. Unlike a normal subquery, it cannot be executed independently because it uses values from the outer query. The correlated subquery is executed once for every row processed by the outer query. It is commonly used to compare values within the same table or between related tables, making it useful for solving complex database queries.

Key Points

1. A correlated subquery is a query nested inside another SQL query.
2. It depends on the outer query for its execution.
3. It executes once for each row returned by the outer query.
4. It is mainly used to compare values within the same table or related tables.
5. It helps identify employees whose salary is greater than the average salary of their own department.
6. It provides accurate row-by-row comparison of data.
7. It is useful for employee performance analysis and salary comparison.
8. Correlated subqueries are widely used in business reports, payroll systems, and financial analysis.
9. Although they may take more execution time than normal subqueries, they provide precise and meaningful results.
10. They improve the flexibility of SQL by handling complex conditions that cannot be solved using simple queries.

Applications

- Employee salary analysis.
- Department-wise performance evaluation.
- Student marks comparison.
- Sales and financial reporting.
- Payroll and HR management systems.

Advantages

- Performs accurate row-by-row comparison.
- Supports complex query conditions.
- Improves data analysis and reporting.
- Retrieves meaningful and relevant information.
- Widely used in real-world database applications.

Q5)

Introduction

A JOIN is an SQL operation used to combine data from two or more related tables based on a common column. It allows users to retrieve meaningful information stored across multiple tables. JOIN operations eliminate data redundancy, improve data retrieval, and help generate accurate reports. Different types of JOINS return different sets of records depending on the relationship between the tables.

Key Points

1. A JOIN is used to combine records from two or more related tables.
2. It retrieves data based on a common attribute such as Employee ID or Department ID.
3. INNER JOIN returns only the records that have matching values in both tables.
4. LEFT OUTER JOIN returns all records from the left table and matching records from the right table.
5. RIGHT OUTER JOIN returns all records from the right table and matching records from the left table.
6. FULL OUTER JOIN returns all matching and non-matching records from both tables.

7. CROSS JOIN returns every possible combination of rows from both tables.
8. JOIN operations reduce data duplication by storing related information in separate tables.
9. They are widely used to generate employee, project, customer, and sales reports.
10. JOINS improve data retrieval, database efficiency, and decision-making.

Applications

- Employee and project management.
- Banking and financial systems.
- Airline reservation systems.
- Hospital management systems.
- E-commerce and inventory management.

Advantages

- Combines related data from multiple tables.
- Reduces data redundancy.
- Improves database performance.
- Generates meaningful reports.
- Supports efficient data analysis and decision-making.

Q6)

A View is a virtual table in a relational database that is created from one or more existing tables. It does not store data physically but displays data retrieved from the base tables whenever it is accessed. Views simplify complex queries, improve database security, and provide users with only the required information. In this case, the DeptSalaryView displays the department name, total number of employees, and the average salary of employees in each department.

Key Points

1. A View is a virtual table created from one or more database tables.
2. It does not store data separately; it displays data from the base tables.
3. DeptSalaryView contains the department name, total employees, and average salary.
4. Views simplify complex database queries by presenting only the required information.
5. They improve database security by restricting access to sensitive data.
6. Views always display the latest data available in the base tables.
7. They reduce query repetition and improve code reusability.
8. Views make report generation faster and easier.
9. They are commonly used in management dashboards and salary reports.
10. Views improve database maintenance, readability, and overall efficiency.

Applications

- Department-wise salary analysis.
- Employee management reports.
- Management dashboards.
- Payroll systems.
- Business reporting and decision-making.

Advantages

- Improves database security.
- Simplifies complex queries.
- Reduces data redundancy.
- Provides consistent and updated information.
- Enhances database performance and maintenance.

Q7)

Relational Algebra is a mathematical query language used in relational database systems to retrieve and manipulate data. It consists of various operations such as Selection, Projection, Join, Union, Difference, and Division. To find the names of students who are enrolled in **all courses offered**, the **Division (÷) operation** is generally used. This operation helps retrieve records that satisfy a universal condition and is useful in solving complex database queries.

Key Points

1. Relational Algebra is the theoretical foundation of SQL.
2. It performs operations on one or more relations (tables).
3. The **Division ($\div$) operation** is used when a query involves "**all**" conditions.
4. It identifies students who are enrolled in every course offered.
5. Division compares two relations and returns only the qualifying records.
6. It is useful for solving universal quantification problems in databases.
7. This operation simplifies complex database retrieval tasks.
8. It is widely used in educational, banking, and business database systems.
9. Relational Algebra operations improve query efficiency and accuracy.
10. They help retrieve meaningful information from relational databases.

Applications

- Finding students enrolled in all courses.
- Identifying employees working on all projects.
- Finding customers who purchased all available products.
- Educational and business database management.
- Complex query processing in DBMS.

Advantages

- Simplifies complex database queries.
- Improves query performance.
- Retrieves accurate and meaningful information.
- Supports efficient database management.
- Forms the basis for SQL query processing.

Q8)

Data Manipulation Language (DML) is a part of SQL used to manage the data stored in database tables. It allows users to add, modify, and remove records while maintaining data accuracy. The main DML operations are **INSERT, UPDATE, and DELETE**. These operations work together with integrity constraints such as **Primary Key, Foreign Key, NOT NULL, and UNIQUE** to ensure that the database remains consistent, valid, and reliable.

Key Points

1. DML stands for **Data Manipulation Language**.
2. It is used to insert, update, and delete records in database tables.
3. **INSERT** is used to add new records into a table.
4. **UPDATE** is used to modify existing records based on specific conditions.
5. **DELETE** is used to remove unwanted records from a table.
6. **Primary Key** ensures that each record is unique and prevents duplicate entries.
7. **Foreign Key** maintains relationships between related tables and ensures referential integrity.
8. **NOT NULL** ensures that mandatory fields always contain valid data.
9. **UNIQUE** prevents duplicate values in selected columns and improves data consistency.
10. DML operations with integrity constraints help maintain accurate, secure, and reliable data in the database.

Applications

- Student information management systems.
- Banking and financial applications.
- Hospital management systems.
- Airline reservation systems.
- Inventory and sales management.

Advantages

- Enables efficient data management.
- Maintains data accuracy and consistency.
- Prevents invalid and duplicate records.
- Supports secure and reliable database transactions.
- Improves overall database performance.

Q8)

Data Manipulation Language (DML) is a category of SQL commands used to manage the data

stored in database tables. It allows users to insert new records, update existing records, and delete unwanted records. While performing these operations, **integrity constraints** ensure that the data remains accurate, valid, and consistent. Common integrity constraints include **Primary Key, Foreign Key, NOT NULL, UNIQUE, and CHECK**, which prevent invalid or duplicate data from being stored in the database.

Key Points

1. DML stands for **Data Manipulation Language** and is used to manage data in database tables.
2. The three main DML operations are **INSERT, UPDATE, and DELETE**.
3. **INSERT** is used to add new records into a table.
4. **UPDATE** is used to modify existing records without affecting other data.
5. **DELETE** is used to remove unwanted or unnecessary records from a table.
6. **Primary Key** ensures that every record is unique and prevents duplicate entries.
7. **Foreign Key** maintains relationships between tables and enforces referential integrity.
8. **NOT NULL, UNIQUE, and CHECK** constraints ensure that only valid and complete data is stored.
9. Integrity constraints help maintain data consistency and prevent errors during DML operations.
10. DML operations are widely used in banking, hospitals, educational institutions, airline reservation systems, and business applications for day-to-day data management.

Applications

- Student Management Systems.
- Banking and Financial Systems.
- Hospital Management Systems.
- Airline Reservation Systems.
- Inventory and Sales Management.
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Advantages

- Enables efficient data insertion, modification, and deletion.
- Maintains data accuracy and consistency.
- Prevents duplicate and invalid records.
- Supports secure and reliable database transactions.
- Improves overall database performance and integrity.

Q9)

Introduction

Tuple Relational Calculus (TRC) is a **non-procedural query language** used in relational database systems. Unlike Relational Algebra, it specifies **what data should be retrieved** rather than **how to retrieve it**. It is based on **mathematical predicate logic** and allows users to define conditions that the required tuples (rows) must satisfy. Tuple Relational Calculus forms the theoretical basis of SQL and is useful for expressing complex database queries.

Key Points

1. Tuple Relational Calculus (TRC) is a non-procedural query language.
2. It is based on mathematical predicate logic.
3. TRC specifies **what information is required**, not the procedure to retrieve it.
4. The database management system determines the best method to execute the query.
5. A tuple represents a single row or record in a relation (table).
6. TRC retrieves only those tuples that satisfy the given conditions.
7. It is useful for expressing complex queries in a simple and logical manner.
8. In this scenario, it identifies employees who belong to the **IT department** and whose **salary is greater than 50000**.
9. TRC improves query flexibility and supports efficient data retrieval.
10. It is widely used in database theory, query optimization, and relational database design.

Applications

- Employee and payroll management.
- Student information systems.
- Banking and financial applications.

- Hospital management systems.
- Business reporting and data analysis.

Advantages

- Easy to express complex conditions.
- Retrieves only the required records.
- Provides flexibility in query formulation.
- Forms the theoretical foundation of SQL.
- Improves database query efficiency.

Q9)

Introduction

An **Airline Reservation System** is a database application used to manage passenger information, flight schedules, ticket bookings, seat allocation, and payment details. It stores all information in related tables to avoid data redundancy and maintain consistency. A well-designed **relational schema** establishes relationships between different entities using primary keys and foreign keys, ensuring efficient storage, retrieval, and management of airline reservation data.

Key Points

1. A relational schema is the logical structure of a database that defines tables and their relationships.
2. The Airline Reservation System consists of **Passenger, Flight, Booking, and Payment** tables.
3. The **Passenger** table stores passenger details such as Passenger ID, Name, Age, Gender, and Contact Number.
4. The **Flight** table stores flight details such as Flight ID, Flight Name, Source, Destination, Departure Time, and Arrival Time.
5. The **Booking** table records ticket bookings and links passengers with flights using unique identifiers.
6. The **Payment** table stores payment details such as Payment ID, Booking ID, Amount, and Payment Status.
7. Primary Keys uniquely identify each record, while Foreign Keys maintain relationships between tables.
8. SQL features such as **JOINS, Sub-Queries, and Views** help retrieve and analyze information efficiently from related tables.
9. The relational schema reduces data redundancy, improves data integrity, and ensures consistency across the database.
10. Airline Reservation Systems are widely used by airlines and travel agencies to provide secure, fast, and reliable booking services.

Applications

- Flight reservation and ticket booking.
- Passenger information management.
- Flight schedule management.
- Payment and transaction processing.
- Airline reporting and business analysis.

Advantages

- Reduces data redundancy.
- Maintains data accuracy and consistency.
- Provides secure storage of passenger information.
- Enables fast retrieval of booking and flight details.
- Improves the efficiency and reliability of airline operations.